

Web Assignment: Crypto Price Ticker

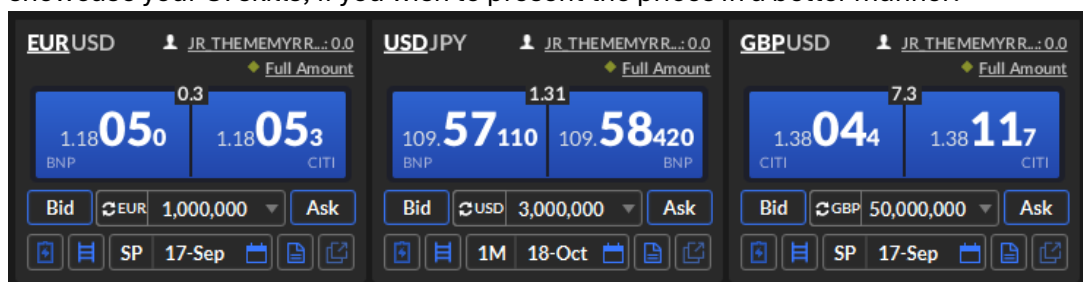
Context

Spark is in the business of FX price data aggregation. In short, this means connecting to various sources, obtaining their price data, and using that information to our customer's benefit. Instead of FX, this assignment is built around Crypto. Crypto's widely available, modern APIs allow you to get started quickly while largely representing as they're found in FX.

Task

Build a client-side price ticker UI that shows BTC/USD, ETH/USD, and other pairs' rates:

- The price ticker should show bid and ask prices for each pair
- Make sure prices update automatically at a timing/interval of your choice
- Use any publicly available API as data source. [Kraken](#) or [Coinbase](#), for instance, both offer public price tickers without authentication.
- You can consider stablecoins such as USDT equivalent to USD in this assignment.
- A user should be able to specify the trade amount, execute on a price (a dummy trade mechanism will do), and see the provider of the price
- Below is a trading screen showing bid and ask prices. Focus on the display of the price, don't get distracted by additional information it contains. This is just a UI reference, do showcase your UI skills, if you wish to present the prices in a better manner.



Technical Considerations

An ideal solution finds scalable and elegant ways to address performance considerations such as:

- Subscribing to multiple instruments from multiple APIs
- Smooth rendering of complex UIs (e.g., high frequency updates, tree structures) under high throughput situations (e.g., 60 instruments on screen with up to 100 updates per second over the network)

Deliverables

- We'd like to see a live demo and want to have a look at the source code before you come by and walk us through your solution. Ideally, you'd be able to provide a deployed version too (e.g. GitHub pages).
- Do write a quick summary of how you approached the task, what your major architectural and design decisions were, and other information that might be relevant (e.g., things you would've done differently if you had more time, diagrams, etc.).